



# Physics Worksheet

## Fluid Mechanics

Name: \_\_\_\_\_ Student #: \_\_\_\_\_ Student Email #: \_\_\_\_\_

1. The following question concerns Bernoulli's principle

(a) What is Bernoulli's principle?

(b) How is the concept of Bernoulli's principle used to explain how lift on an aircraft works?

(c) How can Bernoulli's principle help explain why tornadoes rip roofs off of houses?

2. The following questions concern the buoyancy force

(a) What is the relationship between pressure and depth in water?

(b) As pressure increases with depth of the water, does the buoyancy force also increase? Explain why or why not:

3. Let's say you're filling a pot with water from a faucet. Why is the sound of the water crashing into the bottom of the pot louder if you hold the pot at the bottom of the sink rather than close to the top?

4. If the air above a wing is  $0.5\text{atm}$ , what is the velocity of that air?